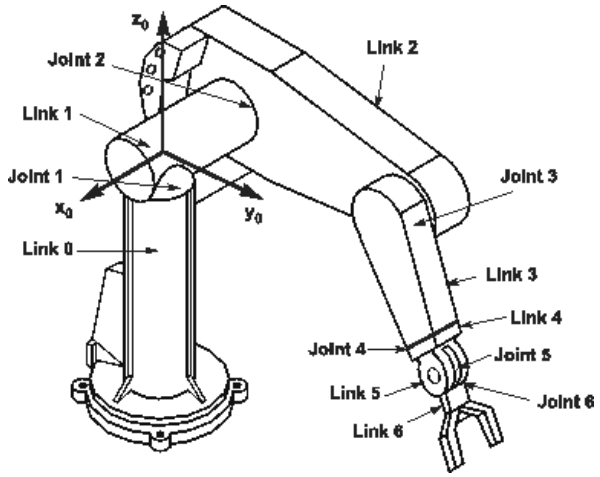
Lecture 2: Translations & Rotations

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What did we learn last time?

What are we doing today?

1. Introduction to Robotics & Recap of Linear Algebra and Mathematical Notation
2. **Translations & Rotation Matrices** ***(Today)***
3. Other Representations for Orientation
4. Transformation Matrices
5. Forward Kinematics & DH Parameters
6. Schematic Representations of Robots, FK from DH, DH Parameters Practice
7. Inverse Kinematics
8. Kinematic Simulation
9. Angular Velocities, Velocity Propagation & Trajectory Generation
10. The Jacobian Matrix, Singularities
11. Analytical & Geometrical Jacobian, Numerical IK, Statics
12. Manipulability, More on the Robotic Systems Toolbox

Topics for Today

Part I:

- **Types of Joints**, **Degrees of Freedom** (DoF)
- **Robot Workspace**
- **Common Robot Frames**

Part II

- **Points** and **Translations** (Vectors)
- **Orientations** and **Rotations** (Matrices)

Part I: Joints, Degrees of Freedom & Common Frames

How did we define a robot last time?

A robot from the POV of Kinematics

A robot is a series of:

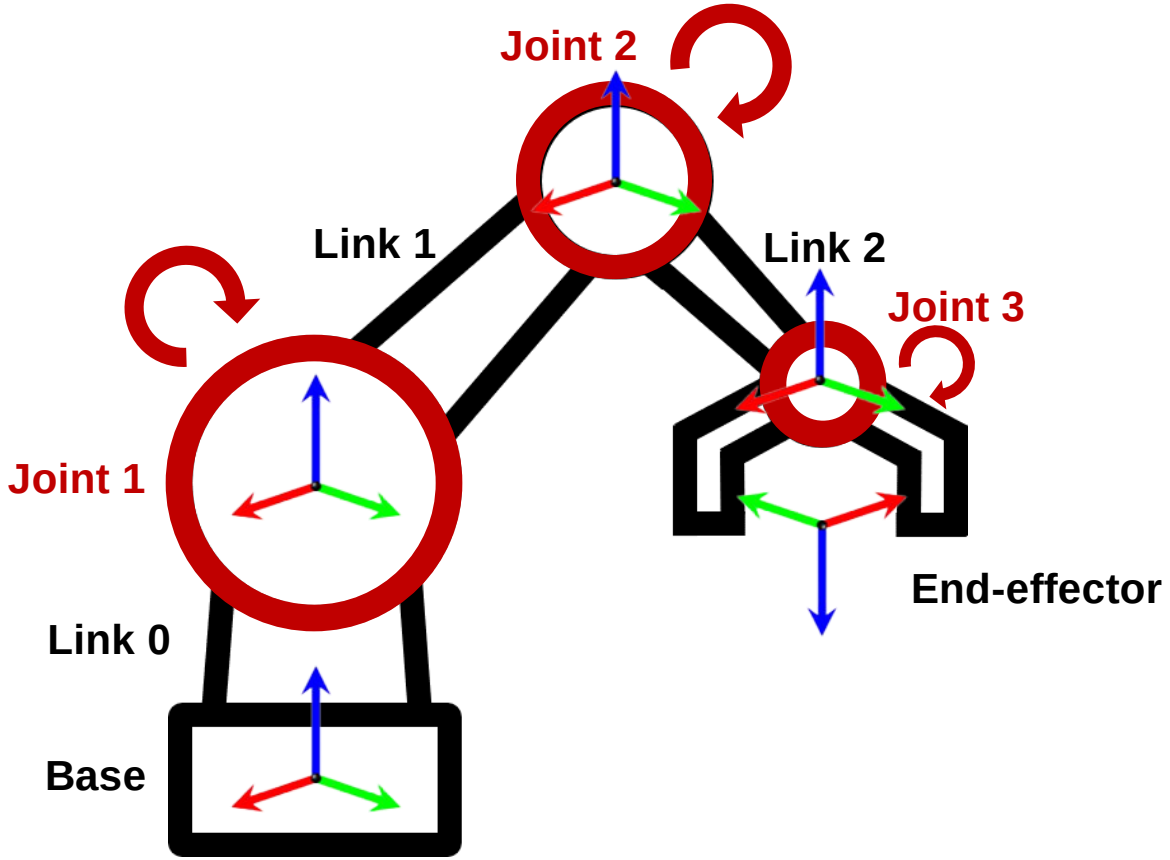
- Rigid **links**, which **do not move** on their own.
- **Joints**, which **allow relative movement** between links.

We can **place coordinate frames at each of the joints.**

This allows us to describe how:

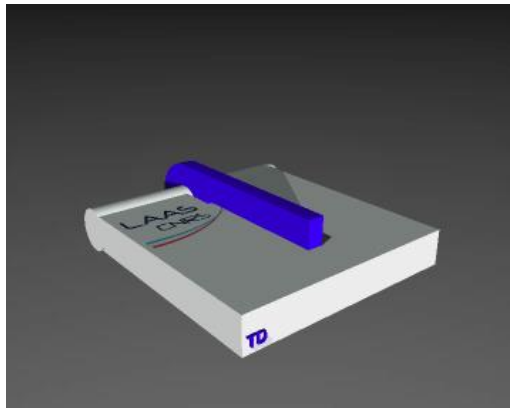
- The **links move with respect to each other**.
- The **end-effector moves relative to the base**.

This representation is called a **Rigid Body Tree**.

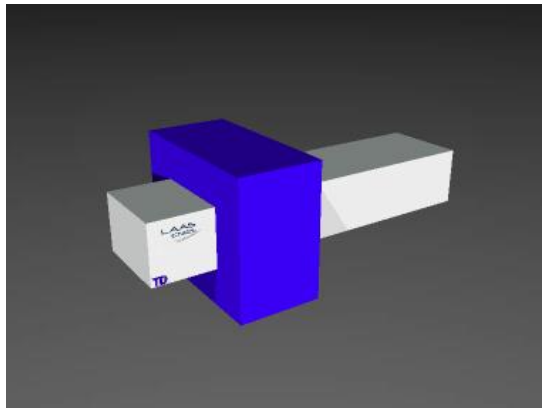


Types of Robot Joints

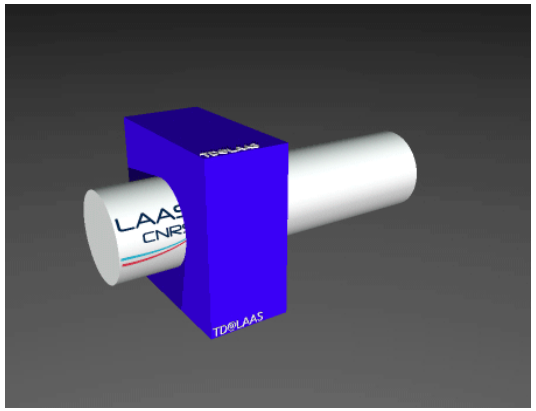
Joints can be of **different types** depending on **how they allow motion**:



Revolute



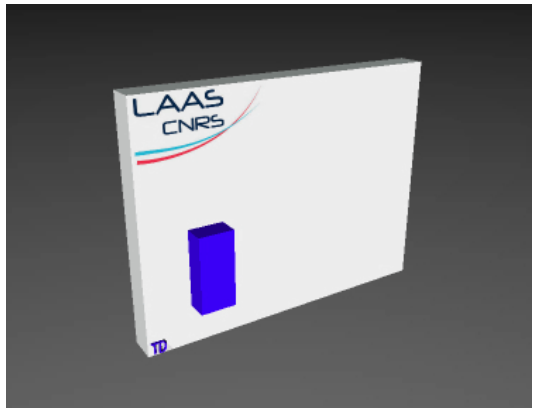
Prismatic



Cylindrical



Spherical



Planar

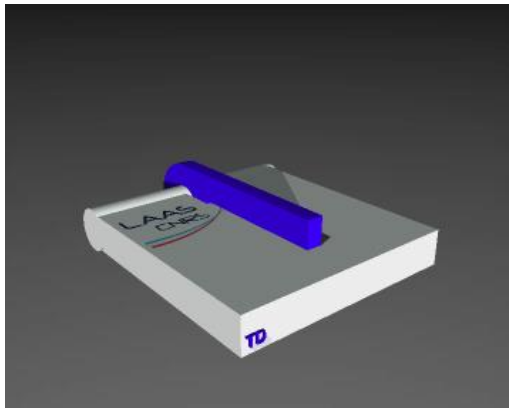
…and others

Source: https://gepettoweb.laas.fr/doc/stack-of-tasks/pinocchio/master/doxygen-html/md_doc_c-maths_c-joints.html

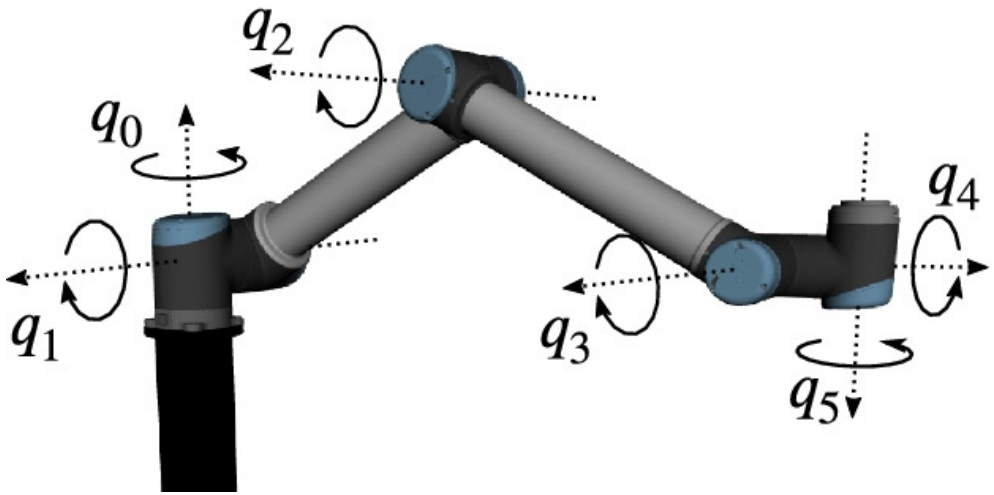
Degrees of Freedom (DoF)

The number of **degrees of freedom** indicates **how many variables need to be specified** in order to fully understand where a mechanism is located.

For example, a **revolute joint has one degree of freedom** (an angle):



How many DOF does each of the joints in the previous slide have?

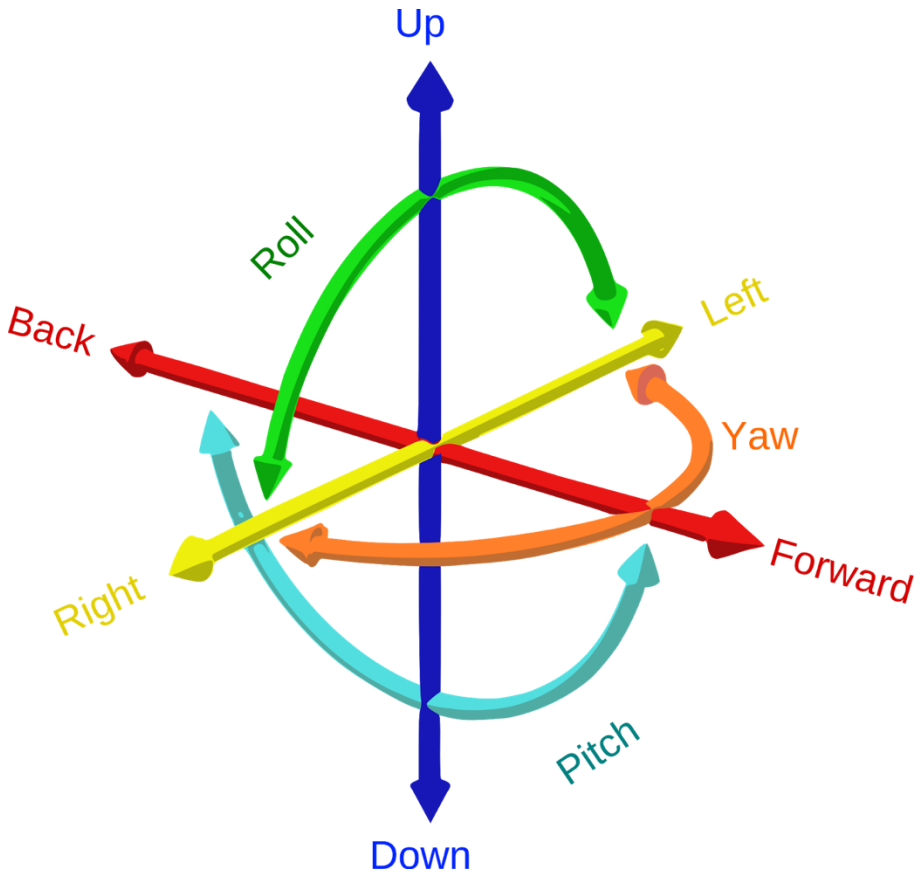


This robot (Universal Robots) consists only of revolute joints.

How many DOF does it then have?

How many DOF does our (3D) world have?





Source: https://prog.world/six-degrees-of-freedom-3d-object-detection-and-more/

How many DOF do the following robots have?



Articulated Robots



SCARA
(Selective Compliance Assembly Robot Arm)



Delta Robots



Cartesian (XYZ)



Mobile Robots
(Automated Guided Vehicles/ Autonomous Mobile Robots)



Drones
(Unmanned Aerial Vehicles)



Humanoids

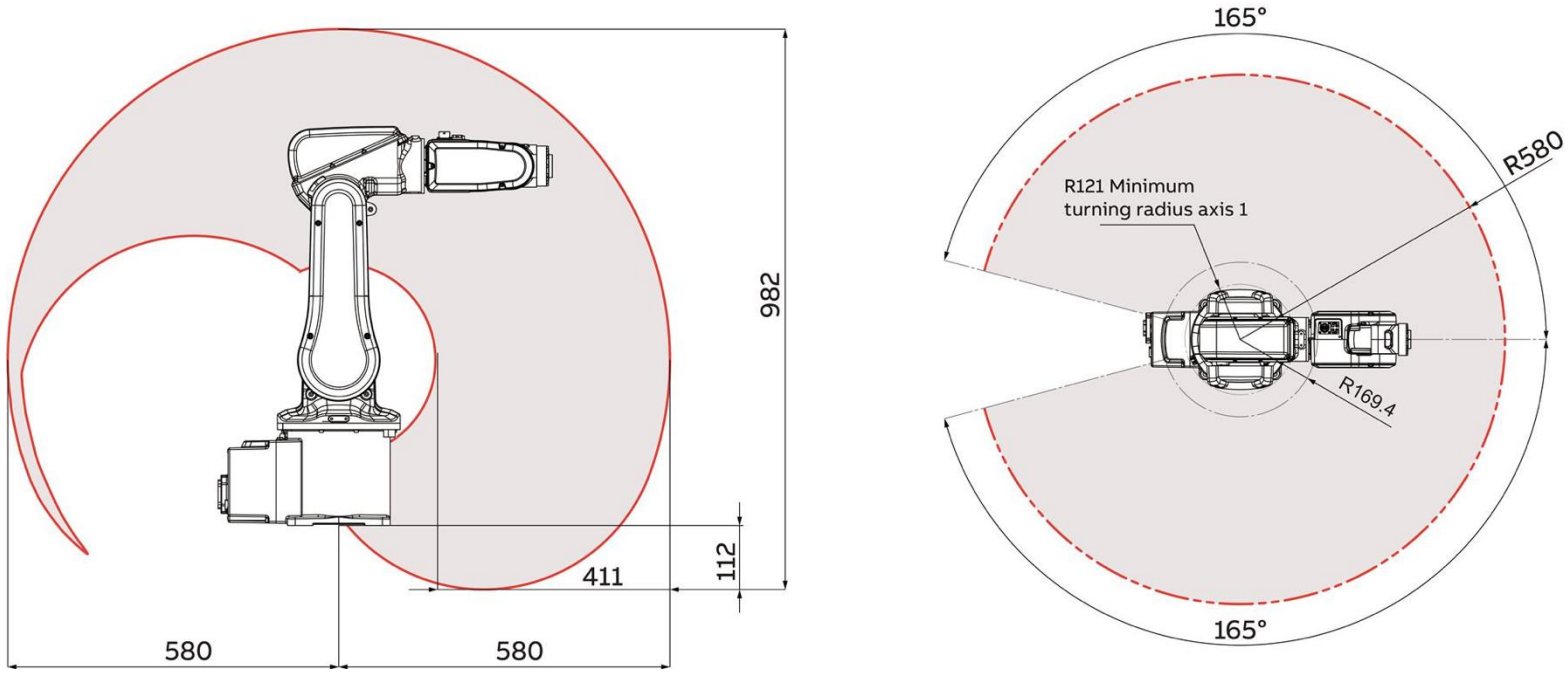
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Robot Workspace

The **workspace** is the **area around** itself that a robot **can reach**.

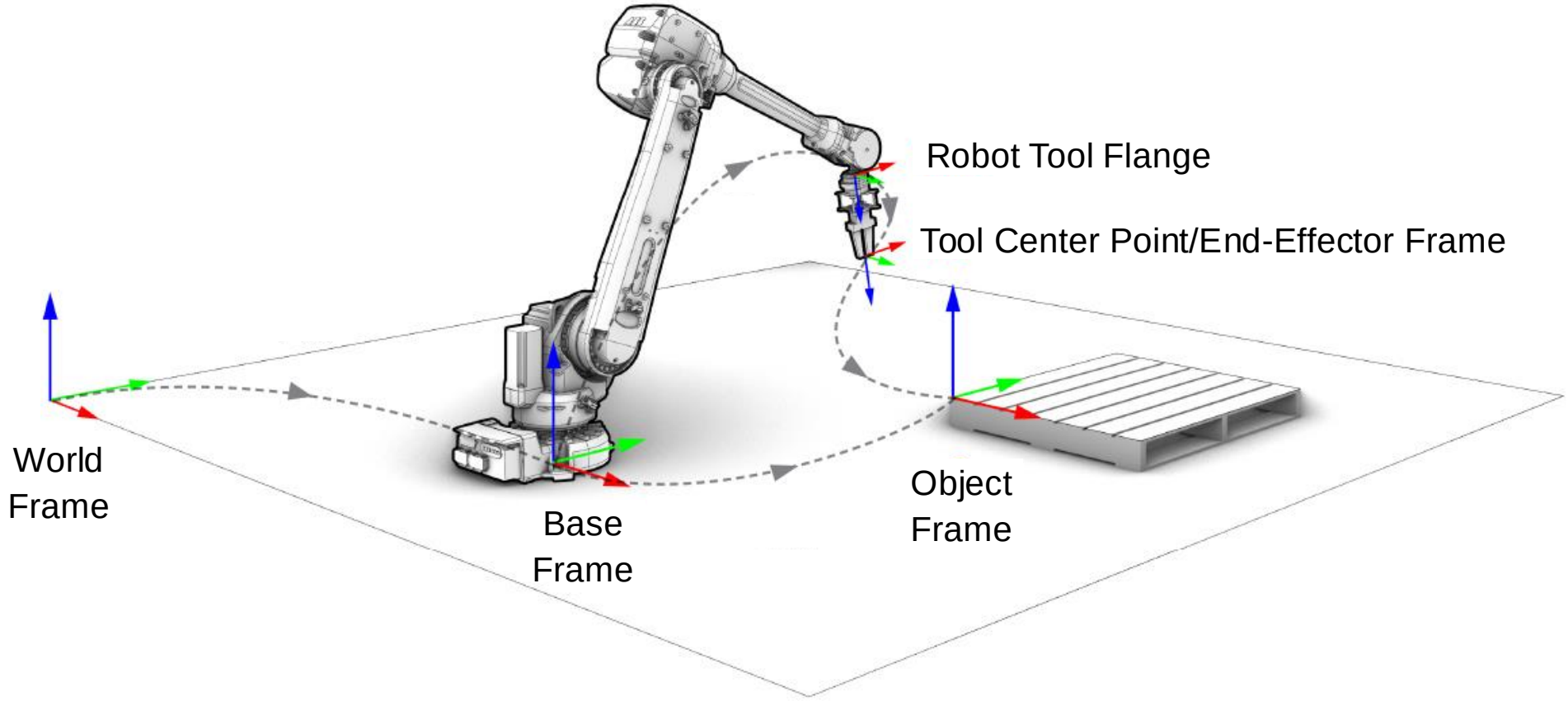
This is **given by** the **mechanical design** of the robot.

Workspace of an ABB IRB120 industrial robot:



Source: http://jderobot.github.io/RoboticsAcademy/exercises/IndustrialRobots/pick_place

Common Robot Frames



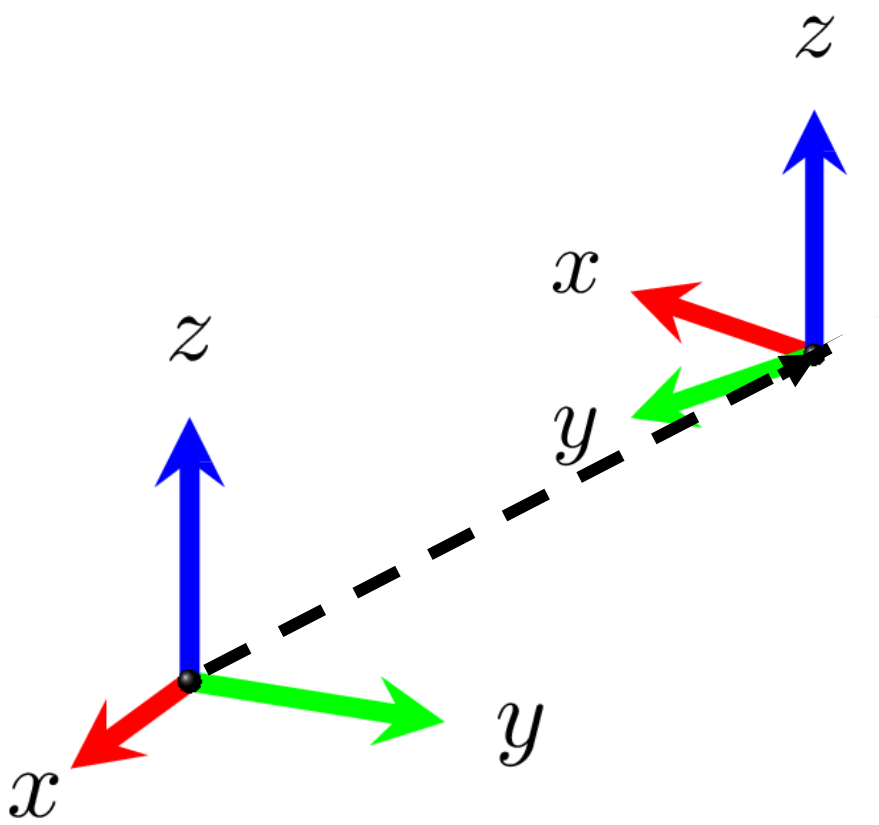
Source: https://gramaziokohler.github.io/compas_fab/latest/examples/01_fundamentals/02_coordinate_frames.html

…and we are back to frames.

Well, this seems to be a recurring theme…

Part II: Translations & Rotations

Translation & Rotation



Recall, that our **frames can:**

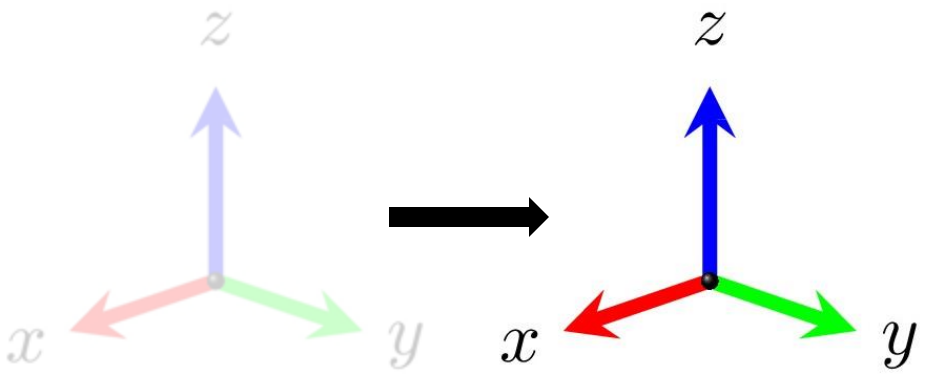
- **Move**
- **Rotate**

…and they can do this **relative to other frames**.

What mathematical tools do we need?

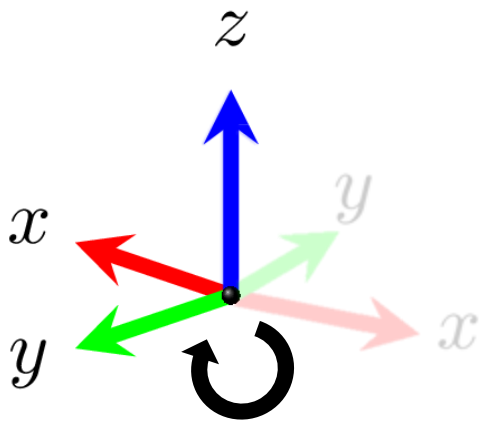
We need to be able to describe:

Difference in **position**
between frames (**translation**)



Vectors

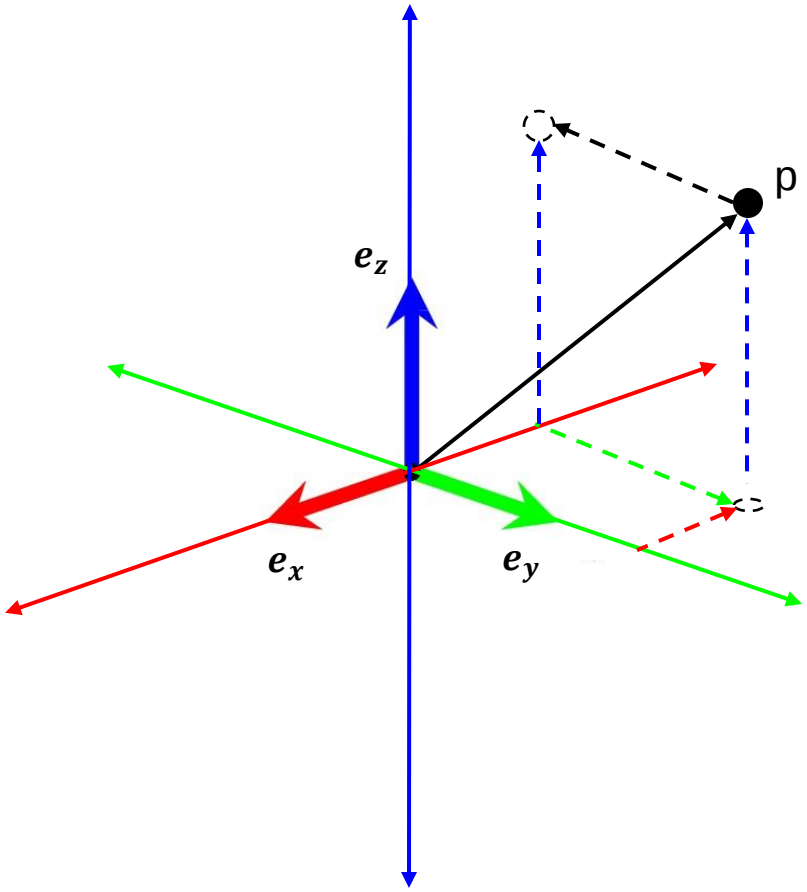
Difference in **orientation**
between frames (**rotation**)



Vectors/Matrices

(For example. There are many ways)

Position (in a reference frame)



Position is given as a **vector.**

This vector can be viewed as a combination of:

- **3 unit vectors**
- **3 magnitudes**

(one for each of the axes)

Position

$$\boldsymbol{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = p_x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + p_y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + p_z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = p_x \, \boldsymbol{e_x} + p_y \boldsymbol{e_y} + p_z \boldsymbol{e_z}$$

$\boldsymbol{p} \in \mathbb{R}^3$
$p_{\{x,y,z\}} \in \mathbb{R}$
$\boldsymbol{e}_{\{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}\}} \in \mathbb{R}^3$

Length

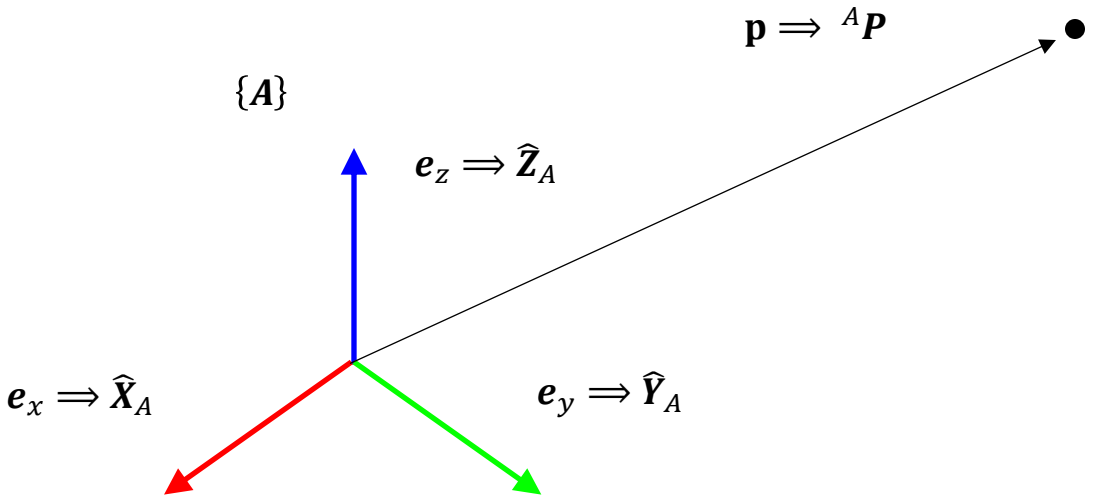
$$|\boldsymbol{p}| = \sqrt{p_x^2 + p_y^2 + p_z^2}$$

$|\boldsymbol{p}| \in \mathbb{R}$

Position (introducing Craig's notation)

In Craig's notation:

- **Frames** are denoted by a **capital letter in curly braces**, e.g. $\{\boldsymbol{A}\}$.
- **Axis unit vectors** are denoted by **capital letters with a hat** and **subscript for the frame**, e.g. $\widehat{\boldsymbol{Z}}_A$.
- **Points** are denoted by a **capital letter** with a **left superscript for the frame**, e.g. ${}^{A}\boldsymbol{P}$.





There is no such thing as
<u>The Golden Notation</u>™ in Robotics*.

Craig's notation is not the only notation.

(We will see some other examples later in the course.)

**Or mathematics in general, for that matter.*

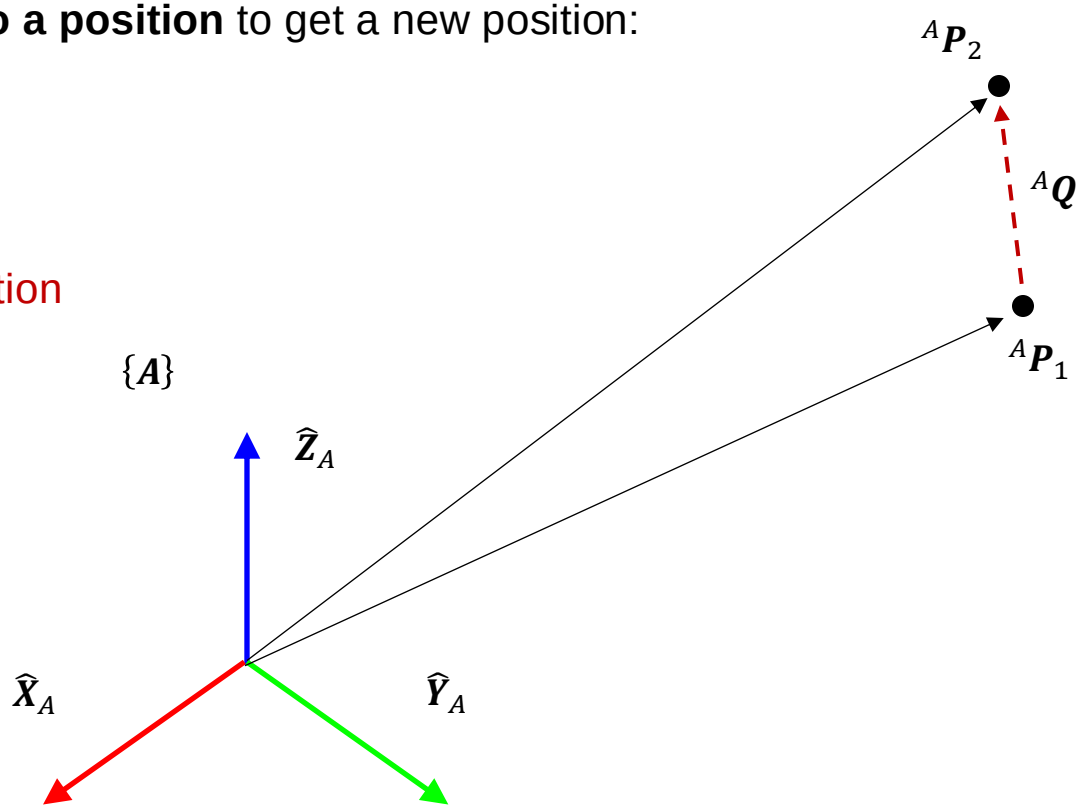
Translation (in a reference frame, {A})

Translation is <u>also</u> given as a **vector.**

We can therefore **add a translation onto a position** to get a new position:

$$^{A}\boldsymbol{P}_2 = {}^{A}\boldsymbol{P}_1 + {}^{A}\boldsymbol{Q}$$

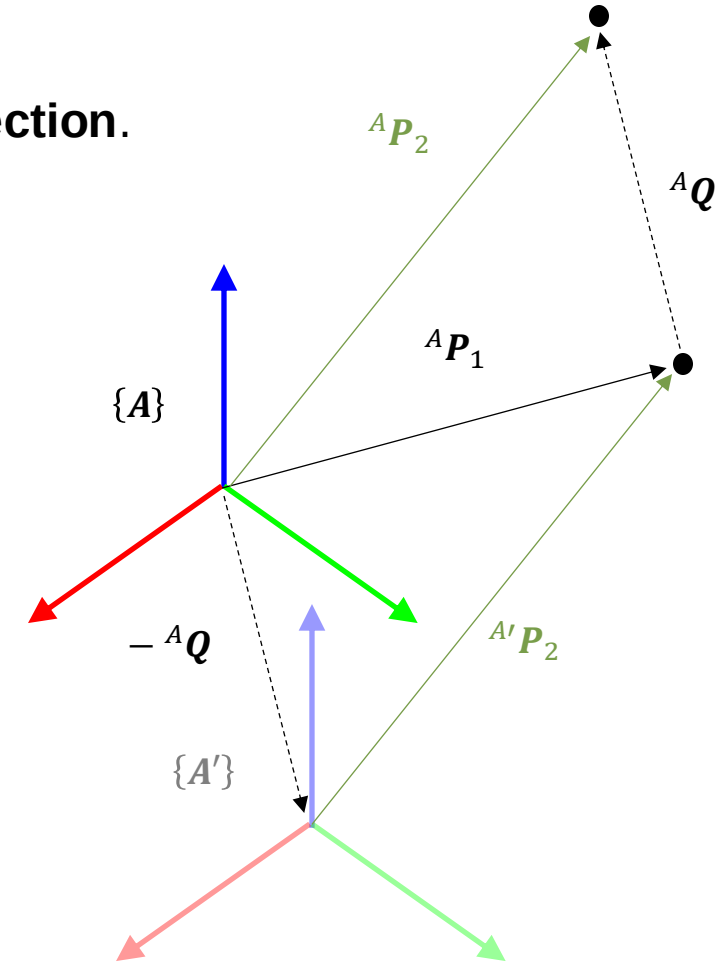
Translation



Translation (viewed as a shifted reference frame {A'})

Translation of a point **can be interpreted**
as **translation of the origin** in the **opposite direction**.

$$^{A'}\boldsymbol{P}_2 = {}^{A}\boldsymbol{P}_2$$



Translation vs. Position (in various reference frames)

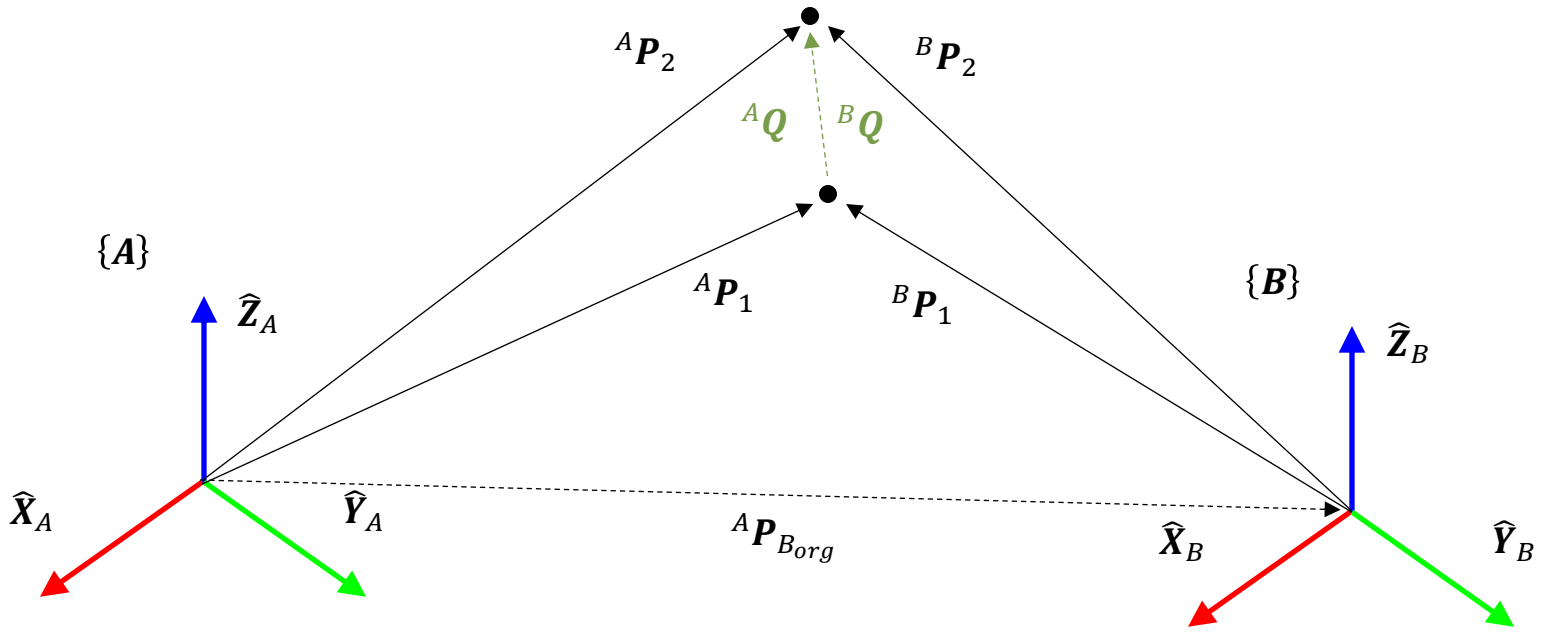
The key difference between position and translation is:

- **Translation** is an **operation;** it is **not specific** to reference frames.
- **Position** is a **description**; it depends on the reference frame.

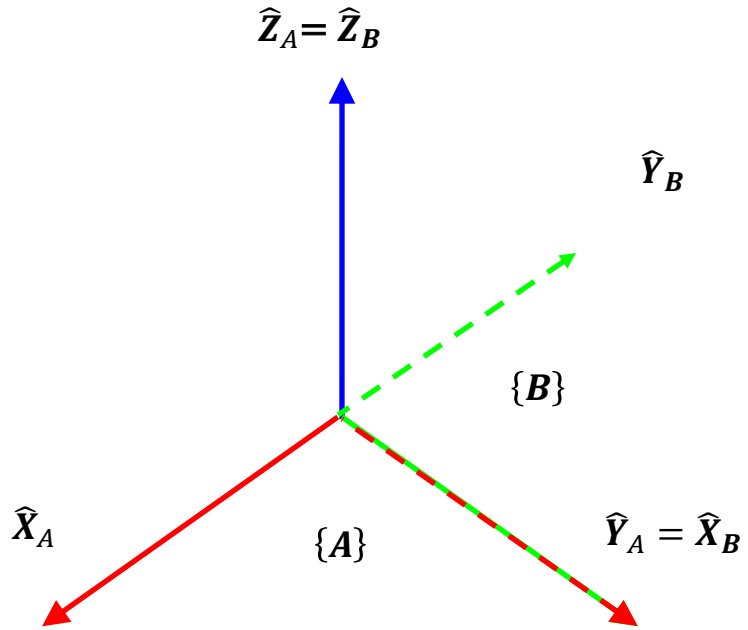
$$
\begin{aligned}
{}^{A}\boldsymbol{P}_2 &= {}^{A}\boldsymbol{P}_1 + {}^{A}\boldsymbol{Q} \\
{}^{B}\boldsymbol{P}_2 &= {}^{B}\boldsymbol{P}_1 + {}^{B}\boldsymbol{Q}
\end{aligned}
$$

$$
\begin{aligned}
{}^{A}\boldsymbol{P}_1 &= {}^{A}\boldsymbol{P}_{B_{org}} + {}^{B}\boldsymbol{P}_1 \\
{}^{A}\boldsymbol{P}_2 &= {}^{A}\boldsymbol{P}_{B_{org}} + {}^{B}\boldsymbol{P}_2
\end{aligned}
$$

$$
\Rightarrow {}^{A}\boldsymbol{Q} = {}^{B}\boldsymbol{Q}
$$



Deriving Rotation (I)



Given that we know:

$$^{A}\hat{X}_A = \begin{bmatrix}1\\0\\0\end{bmatrix}, ^{A}\hat{Y}_A = \begin{bmatrix}0\\1\\0\end{bmatrix}, ^{A}\hat{Z}_A = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

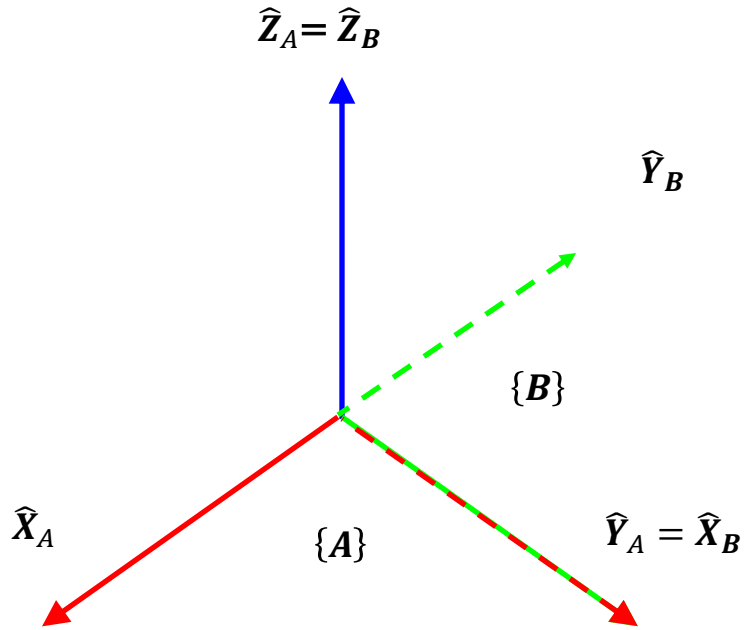
$$^{B}\hat{X}_B = \begin{bmatrix}1\\0\\0\end{bmatrix}, ^{B}\hat{Y}_B = \begin{bmatrix}0\\1\\0\end{bmatrix}, ^{B}\hat{Z}_B = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

What are:

- The axes of {B} seen from {A}: $^{A}\hat{X}_B, ^{A}\hat{Y}_B, ^{A}\hat{Z}_B$?
- The axes of {A} seen from {B}: $^{B}\hat{X}_A, ^{B}\hat{Y}_A, ^{B}\hat{Z}_A$?

Discuss in small groups.

Deriving Rotation (II)

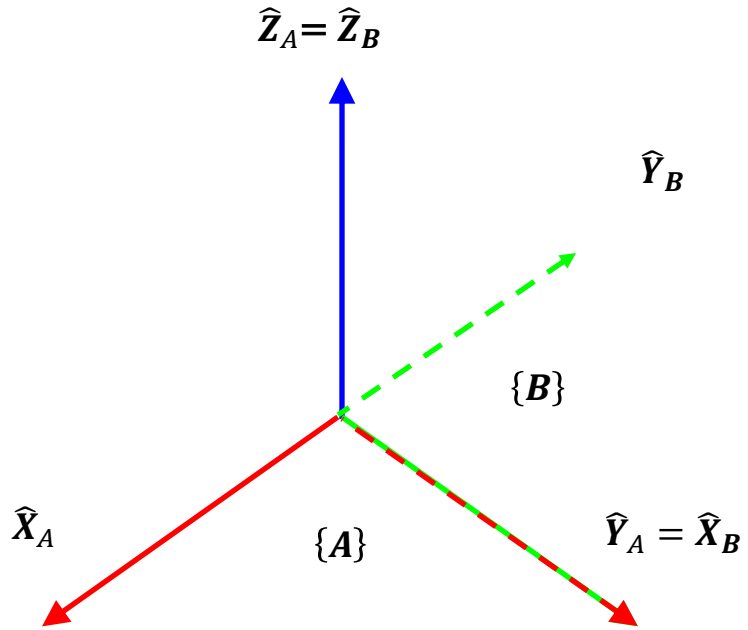


Answer:

$$^{A}\hat{X}_B = \begin{bmatrix}0\\1\\0\end{bmatrix}, ^{A}\hat{Y}_B = \begin{bmatrix}-1\\0\\0\end{bmatrix}, ^{A}\hat{Z}_B = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$^{B}\hat{X}_A = \begin{bmatrix}0\\-1\\0\end{bmatrix}, ^{B}\hat{Y}_A = \begin{bmatrix}1\\0\\0\end{bmatrix}, ^{B}\hat{Z}_A = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

Deriving Rotation (III)



$$^{A}\hat{X}_B = \begin{bmatrix}0\\1\\0\end{bmatrix}, ^{A}\hat{Y}_B = \begin{bmatrix}-1\\0\\0\end{bmatrix}, ^{A}\hat{Z}_B = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$^{B}\hat{X}_A = \begin{bmatrix}0\\-1\\0\end{bmatrix}, ^{B}\hat{Y}_A = \begin{bmatrix}1\\0\\0\end{bmatrix}, ^{B}\hat{Z}_A = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

Let's put it together in a matrix:

- The **columns** describe the axes of {B} in reference frame {A},
- The **rows** describe the axes of {A} in rotated frame {B}.

$$^{A}_{B}\boldsymbol{R} = [^{A}\hat{X}_B \quad ^{A}\hat{Y}_B \quad ^{A}\hat{Z}_B] = \begin{bmatrix}^{B}\hat{X}_A^T\\ ^{B}\hat{Y}_A^T\\ ^{B}\hat{Z}_A^T\end{bmatrix}$$

$$^{A}_{B}\boldsymbol{R} = \begin{bmatrix}0 & -1 & 0\\1 & 0 & 0\\0 & 0 & 1\end{bmatrix}$$

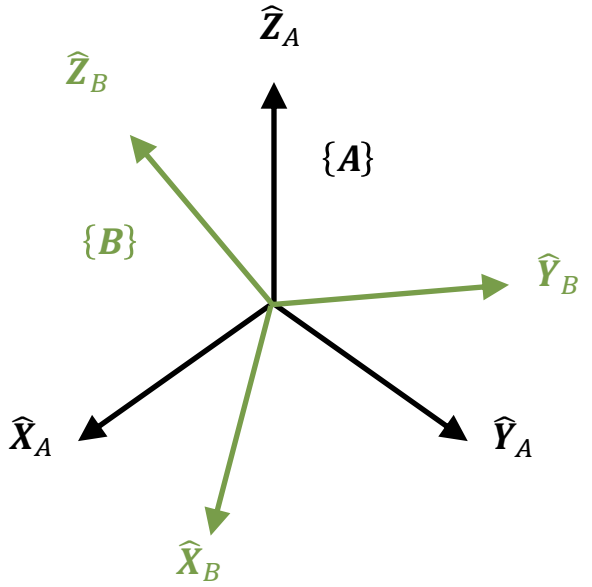
You just constructed your first rotation matrix!

Rotation Matrices

Rotation is an **operation** described by a **rotation matrix** in $\mathbb{R}^{3\times3}$.

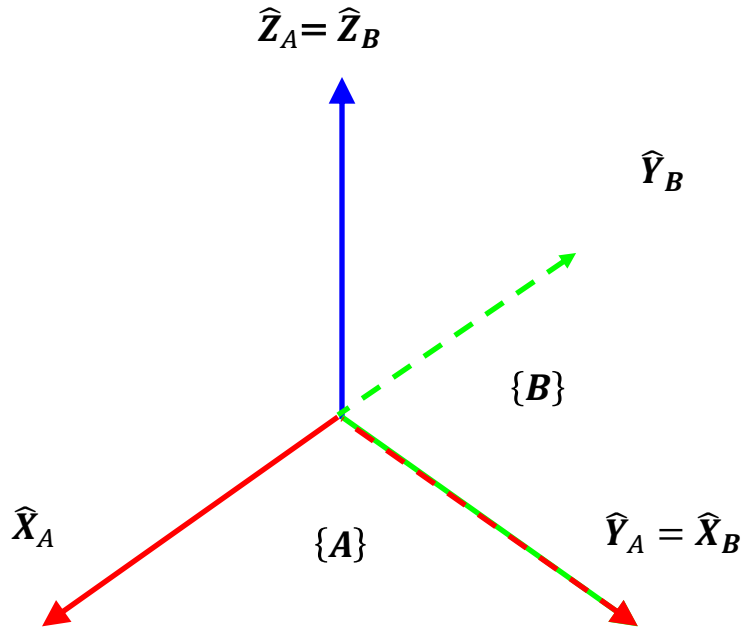
$$ {}^{A}_{B}\boldsymbol{R} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

Note: You can read ${}^{A}_{B}\boldsymbol{R}$ *as "the orientation of {B} seen in {A}"*
OR "the rotation from {A} to {B}".



The **columns** describe the axes of {B} in reference frame {A},
The **rows** describe the axes of {A} in rotated frame {B}.

$$ {}^{A}_{B}\boldsymbol{R} = [{}^{A}\widehat{\boldsymbol{X}}_B \quad {}^{A}\widehat{\boldsymbol{Y}}_B \quad {}^{A}\widehat{\boldsymbol{Z}}_B] = \begin{bmatrix} {}^{B}\widehat{\boldsymbol{X}}_A^T \\ {}^{B}\widehat{\boldsymbol{Y}}_A^T \\ {}^{B}\widehat{\boldsymbol{Z}}_A^T \end{bmatrix} $$



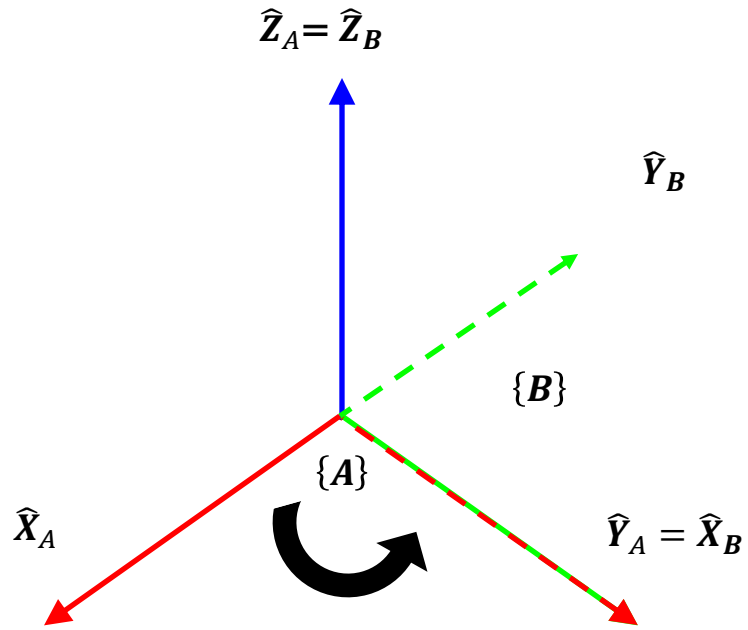
How many degrees and around which axis did we rotate?

Answer: 90° *around the z-axis*

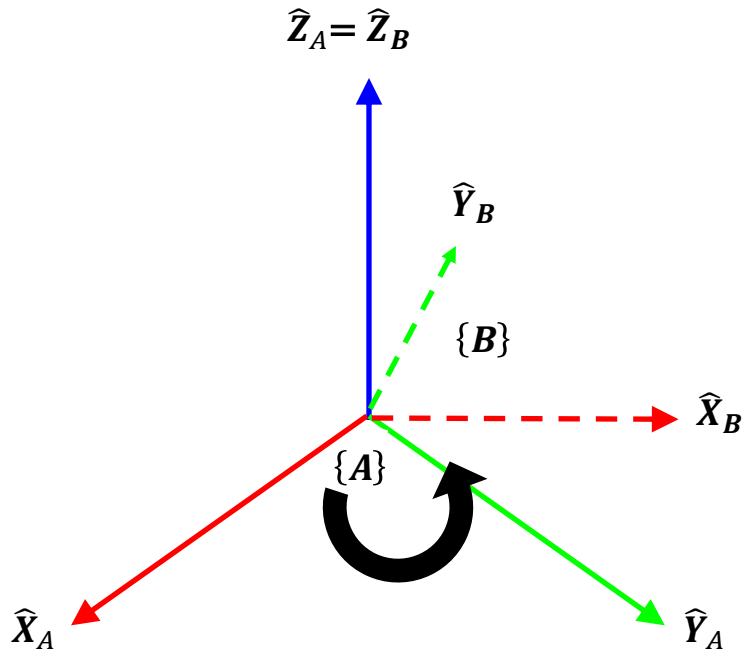
Rotation Matrix (around the *z*-axis)

Let us stick to z-axis rotations…

The previous example is all good, but… **what if it is not so simple**?



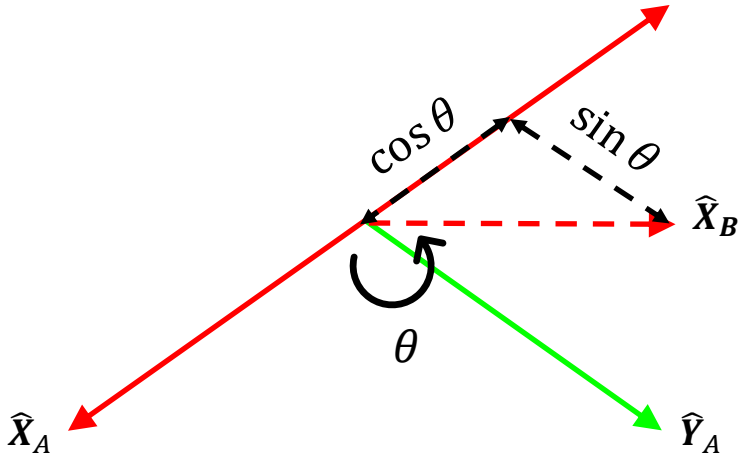
Angle: 90°

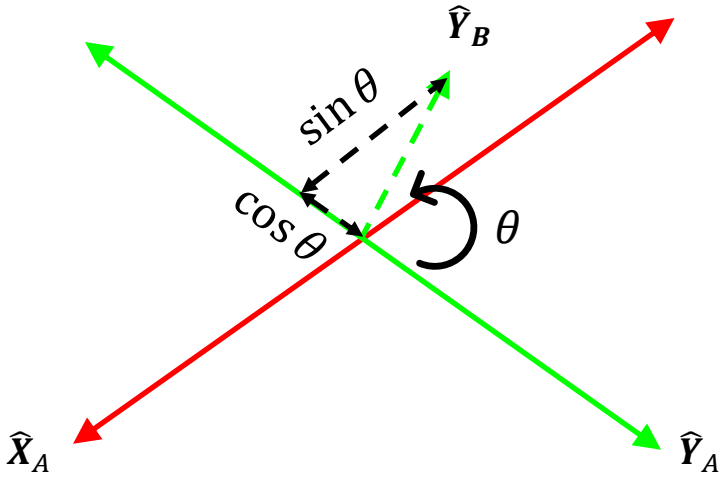


Angle: θ

Rotation Matrix (around the *z*-axis, arbitrary angle)

Simplifying the problem:





$$\boldsymbol{R}_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^A\hat{\boldsymbol{X}}_B = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix}$$

$${}^A\hat{\boldsymbol{Y}}_B = \begin{bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{bmatrix}$$

Rotation Matrices: Properties

Columns & rows are mutually **perpendicular:**

$$
\begin{aligned}
{}^{A}\hat{\boldsymbol{X}}_B \times {}^{A}\hat{\boldsymbol{Y}}_B &= {}^{A}\hat{\boldsymbol{Z}}_B \qquad & {}^{A}\hat{\boldsymbol{X}}_B \cdot {}^{A}\hat{\boldsymbol{Y}}_B &= 0 \\
{}^{A}\hat{\boldsymbol{Y}}_B \times {}^{A}\hat{\boldsymbol{Z}}_B &= {}^{A}\hat{\boldsymbol{X}}_B & {}^{A}\hat{\boldsymbol{Y}}_B \cdot {}^{A}\hat{\boldsymbol{Z}}_B &= 0 \\
{}^{A}\hat{\boldsymbol{Z}}_B \times {}^{A}\hat{\boldsymbol{X}}_B &= {}^{A}\hat{\boldsymbol{Y}}_B & {}^{A}\hat{\boldsymbol{Z}}_B \cdot {}^{A}\hat{\boldsymbol{X}}_B &= 0
\end{aligned}
$$

➡ ${}^{A}_{B}\boldsymbol{R}$ is **orthogonal**

Columns & rows are **unit vectors:**

$$\left|{}^{A}\hat{\boldsymbol{X}}_B\right| = \left|{}^{A}\hat{\boldsymbol{Y}}_B\right| = \left|{}^{A}\hat{\boldsymbol{Z}}_B\right| = 1$$

$$\left|{}^{B}\hat{\boldsymbol{X}}_A^T\right| = \left|{}^{B}\hat{\boldsymbol{Y}}_A^T\right| = \left|{}^{B}\hat{\boldsymbol{Z}}_A^T\right| = 1$$

➡ ${}^{A}_{B}\boldsymbol{R}$ is **orthonormal**

What does this imply?

$${}^{A}_{B}\boldsymbol{R}^{-1} = {}^{B}_{A}\boldsymbol{R}$$

$${}^{A}_{B}\boldsymbol{R}^{T} = {}^{B}_{A}\boldsymbol{R}$$

➡ $\boxed{{}^{A}_{B}\boldsymbol{R}^{-1} = {}^{A}_{B}\boldsymbol{R}^{T}}$

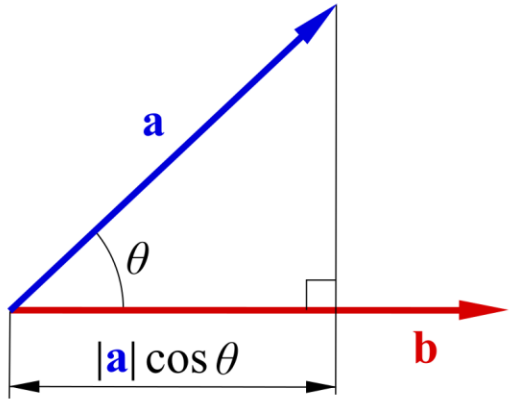
This is very important for computational reasons!

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Points in Frames (as projections onto unit vectors)

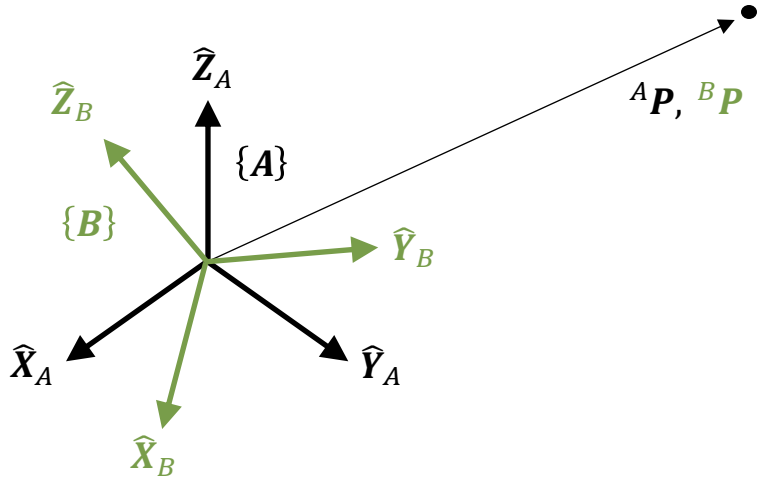
From the previous lecture, remember that the **dot product can be interpreted as a projection**:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos(\theta)$$



We can view the **elements of a vector** as a **projection onto the unit vectors of the frame** it is represented in:

$$
\begin{aligned}
{}^{A}px &= {}^{B}\hat{\boldsymbol{X}}_{A} \cdot {}^{B}\boldsymbol{P} \\
{}^{A}py &= {}^{B}\hat{\boldsymbol{Y}}_{A} \cdot {}^{B}\boldsymbol{P} \\
{}^{A}pz &= {}^{B}\hat{\boldsymbol{Z}}_{A} \cdot {}^{B}\boldsymbol{P}
\end{aligned}
$$



Rotating Points (& Understanding Notation)

We have seen that **we can describe** the **rotation from one frame to another** through a **rotation matrix**.

We can also **apply rotations to points**.

We can use **notation** to determine if a rotation is possible:

- **Subindices on the left cross with superindinces on the right:**

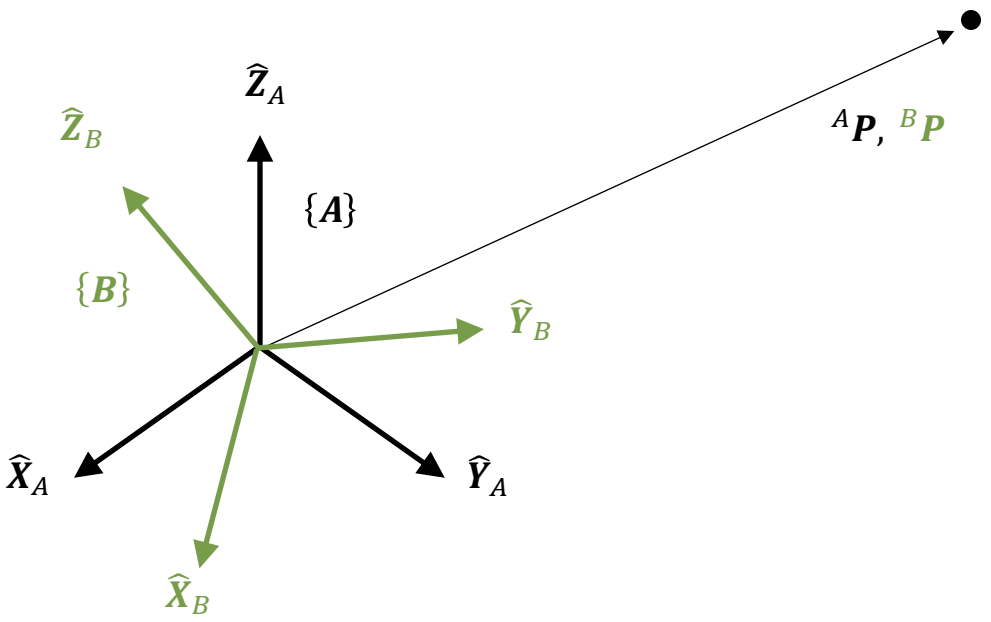
$$ {}^{A}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}\,{}^{B}\boldsymbol{P} $$

- **Otherwise, the operation is not possible:**

$$ {}^{A}\boldsymbol{P} = {}^{B}_{A}\boldsymbol{R}\,{}^{B}\boldsymbol{P} $$

- **However, it is also possible to use the inverse:**

$$ {}^{B}\boldsymbol{P} = {}^{A}_{B}\boldsymbol{R}^{-1}\,{}^{A}\boldsymbol{P} = {}^{B}_{A}\boldsymbol{R}\,{}^{A}\boldsymbol{P} $$



Chains of Rotations

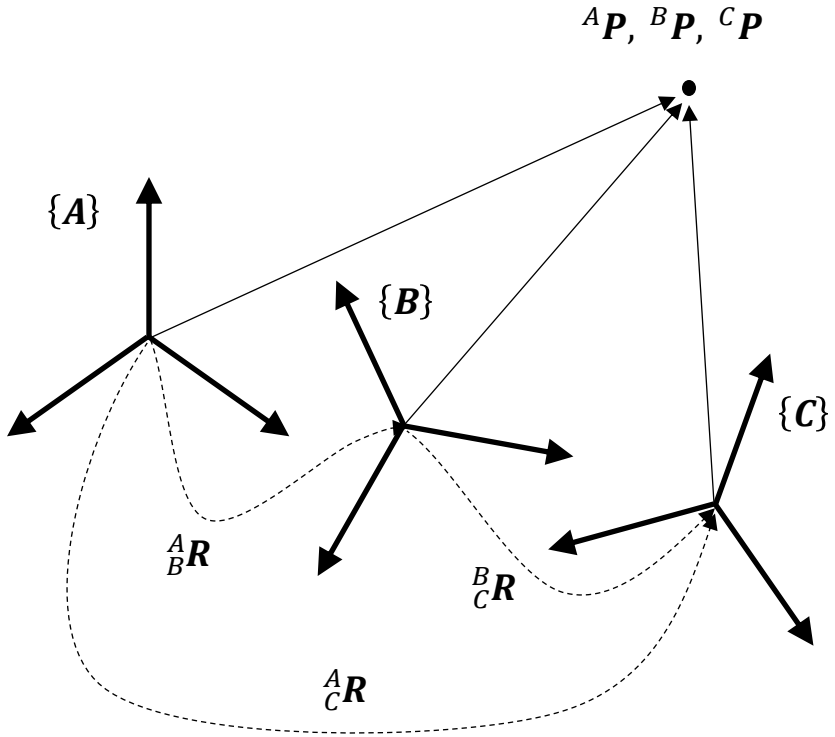
We can chain **multiple rotations together.**

Remember the notation rules from the previous slide.

$$
\begin{aligned}
{}^{A}\boldsymbol{P} &= {}^{A}_{B}\boldsymbol{R}\ {}^{B}\boldsymbol{P} \\
{}^{B}\boldsymbol{P} &= {}^{B}_{C}\boldsymbol{R}\ {}^{C}\boldsymbol{P} \\
{}^{A}\boldsymbol{P} &= {}^{A}_{B}\boldsymbol{R}\ {}^{B}_{C}\boldsymbol{R}\ {}^{C}\boldsymbol{P} = {}^{A}_{C}\boldsymbol{R}\ {}^{C}\boldsymbol{P}
\end{aligned}
$$

$$
\begin{aligned}
{}^{B}\boldsymbol{P} &= {}^{A}_{B}\boldsymbol{R}^{-1}\ {}^{A}\boldsymbol{P} \\
{}^{C}\boldsymbol{P} &= {}^{B}_{C}\boldsymbol{R}^{-1}\ {}^{B}\boldsymbol{P} \\
{}^{C}\boldsymbol{P} &= {}^{B}_{C}\boldsymbol{R}^{-1}\ {}^{A}_{B}\boldsymbol{R}^{-1}\ {}^{A}\boldsymbol{P} = {}^{A}_{C}\boldsymbol{R}^{-1}{}^{A}\boldsymbol{P}
\end{aligned}
$$

$$
\left({}^{A}_{B}\boldsymbol{R}\ {}^{B}_{C}\boldsymbol{R}\right)^{-1} = {}^{B}_{C}\boldsymbol{R}^{-1}\ {}^{A}_{B}\boldsymbol{R}^{-1}
$$



Recap: What have we discussed today

- Some different **kinds of robot joints**.
- The concepts of **degrees of freedom** and **robot workspace.**
- Some **common robot frames.**
- We can use **vectors to describe points.**
  - We can also use **vectors to describe translation** (movement between points).
- We can use **matrices to describe orientations and rotations.**
  - A **rotation matrix** encodes the axes of a frame as seen from the point of view of another frame.
  - **Rotation matrices** are **orthogonal**.
  - We can describe **rotations around one axis** as a rotation matrix that **depends on the sine and cosine of the angle**.
  - We can **chain multiple translations/rotations.**

Take home message:

We can describe the **change in position** between two frames as a **translation vector** and the **change in orientation** as a **rotation matrix.**

We can **chain translations.**

We can **chain rotations**.

SDU ROBOTICS

Thank you for today.

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